Anal. Chem. **2009,** *81,* 9802–9806

Development of a Differential Interference Contrast Thermal Lens Microscope for Sensitive Individual Nanoparticle Detection in Liquid

Hisashi Shimizu, Kazuma Mawatari, and Takehiko Kitamori*

Department of Applied Chemistry, School of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo, Tokyo 113-8656, Japan

A thermal lens microscope (TLM) with a new principle was developed to improve the detection limit of conventional TLM. The detection limit was decreased by introducing a differential interference contrast (DIC) method which realizes background-free photodetection. The new differential interference contrast thermal lens microscope (DIC-TLM) exploits phase contrast resulting from a photothermal effect instead of refraction used in conventional TLM. In order to produce high phase contrast, we fabricated a pair of DIC prisms with a large shear value of 5 μm which is in accordance with the thermal diffusion length. First, we verified the principle of DIC-TLM. The background of TLM measurement was reduced to 1/100 by differential interference, and the signal-to-background (S/B) ratio was improved by 1 order of magnitude. The signal was confirmed to originate from phase contrast, and the expansion of the shear value was effective. Furthermore, we demonstrated counting of individual gold nanoparticles (5 nm) using DIC-TLM. The particles were counted with high signal-to-noise (S/N) ratio, and the S/N ratio was improved by 1 order of magnitude. Finally, we discuss the possibility of single molecule counting in a liquid.

In this decade, the scale of analysis has been rapidly miniaturized, as represented by single cells and micro/nano chips.[1,2] Analysis in micro- and nanospace has various merits such as small sample volumes, short analysis time, and low costs, whereas detection becomes quite difficult. One reason for this is that the number of analyte molecules decreases due to extremely small detection volumes. For example, provided a 1 nM solution is contained in a 1 μm cube, there are 0.6 molecules on average in the cube, which means that single molecule detection is essential for analysis in micro- and nanospace. The most popular method for single molecule detection is laser induced fluorescence (LIF). Single molecule detection by LIF was performed for the first time in 1990,[3,4] and the method has been widely applied for analyses on microchips.[5] However, LIF has the disadvantage that it can detect only fluorescent molecules. As most molecules are nonfluorescent, detection methods for these molecules are required for further applications of microchips. In particular, troublesome fluorescent labeling is necessary for the detection of nonfluorescent proteins and DNAs, and precise labeling of a single molecule is quite difficult. Hence, detection of nonfluorescent molecules will contribute to future bioanalyses on micro/nano chips.

One of the most promising methods of nonlabeled detection is thermal lens spectrometry (TLS).[6,7] TLS is a kind of photothermal spectroscopy that measures absorption and thermal relaxation and has been reported to be more sensitive than absorption spectrometry.[8−11] By combining TLS with a microscope, we have developed a thermal lens microscope (TLM)[12,13] and applied it to various analytical systems on microchips.[14−18] Notably, we succeeded in the determination of a concentration corresponding to 0.4 molecules in a detection volume of 7 fL.[19] In addition, we developed a new method to detect individual particles moving in a liquid for the first time and succeeded in counting metallic nanoparticles one by one.[20] However, counting of individual nonfluorescent molecules has not been achieved so far.

* To whom all correspondence should be addressed: E-mail: kitamori@icl.t.u-tokyo.ac.jp. Fax: +81-3-5841-6039.

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10.1021/ac901888u CCC: $40.75
Published on Web 11/06/2009

This is attributed to the high detection limit of conventional TLM resulting from a high background in photodetection. This background does not originate from the solvent or fluorescent sources but is an effect of the detection principle applied in a TLM.[12] In a conventional TLM, an excitation beam is intensity modulated at around 1 kHz and tightly focused on an analyte solution in a microchip. Following absorption and thermal relaxation of the analyte, the refractive index of the solution changes and a thermal lens effect is induced. Subsequently, a probe beam is refracted (converged or diverged) by the thermal lens effect, and the intensity of the probe beam transmitted through a pinhole changes slightly. Although this change of the probe beam is the signal in conventional TLM, its signal-to-backgound (S/B) ratio is only in the order of 10^{-3}. Therefore, a lock-in amplifier is used for signal recovery. The lock-in amplifier removes a large DC component of the background. However, it cannot remove the low frequency fluctuation around 1 kHz caused by the background. The effect of this fluctuation is greater than the background signals originating from the solvent, fluorescence, or other sources. In other words, the background of the photodetection determines the detection limit of the conventional TLM.

For this reason, a new principle realizing background-free photodetection is required to improve the detection limit. Some background-free methods such as dark field, phase contrast, and differential interference contrast (DIC) are well-known as microscope observation modes. In particular, photothermal spectroscopy using DIC has been implemented for detecting individual nanoparticles fixed in a solid or polymer phase,[21,22] though there is no example of the detection in a liquid. The difficulty of the detection in a liquid is due to the rapid Brownian motion of the particles and short residence time (several ms) in the focus area of the excitation beam (scale around 1 μm). In fact, signal integration to improve the detection limit is restricted. Furthermore, the potential of DIC for background-free photothermal spectroscopy has not been discussed.

In this paper, we introduce the DIC method into TLM and show the development of a differential interference contrast thermal lens microscope (DIC-TLM). In DIC-TLM, the probe beam is separated and interfered by a pair of DIC prisms, which realizes background-free photodetection. The change of refractive index induced by the excitation beam is detected through phase contrast between the two probe beams. In order to produce high phase contrast in a liquid, the distance between the two probe beam spots (shear value) is expanded to 5 μm, considering the thermal diffusion length. First, we verify the principle with respect to the background-free detection, phase contrast, and thermal diffusion length. Afterward, we demonstrate the counting of gold nanoparticles (5 nm in diameter) and compare the performance to conventional TLM. Finally, we discuss the possibility of single molecule counting using DIC-TLM.

EXPERIMENTAL SECTION

Principle of DIC-TLM. Figure 1a shows the principle of DIC-TLM. The probe beam is separated by a DIC prism into two beams with perpendicular polarization. On the other hand, the excitation beam is not separated, as its polarization plane is rotated at an angle of 45 deg. The excitation beam is absorbed by an analyte, and a thermal lens effect is induced. Then, phase contrast appears between the two probe beams due to the difference in refractive index. In the next step, the probe beams are combined again by another DIC prism, which results in a new polarization component. Finally, only this new component is detected as signal by removing the initial polarization component using a polarization filter. When no analyte is in the focus area of the excitation beam, the intensity of the transmitted probe beam is zero. Thus, the background-free photodetection is achieved. In this principle, the relationship between the size of the thermal lens and the distance between the two probe beams is important. The radius of the thermal lens is assumed to be one thermal diffusion length. The thermal diffusion length μ is expressed as

$$\mu = \sqrt{\kappa/\pi C\rho f} \qquad (1)$$

where κ is the thermal conductivity, C is the specific heat, ρ is the density of the solvent, and f is the modulation frequency. The value of μ is 7 μm when f is 1 kHz and water is applied as solvent. The distance between the two probe beam spots, which is called the shear value, is determined by the design of the DIC prisms. A typical shear value of DIC prisms for a commercial microscope

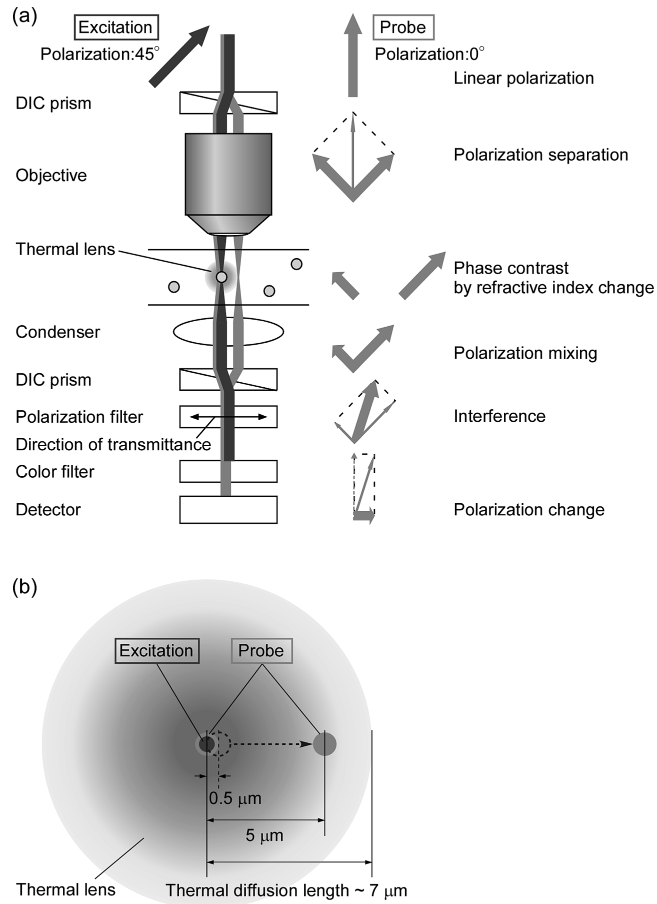

Figure 1. (a) Principle of DIC-TLM. (b) Configuration of beam spots and thermal lens in the focal plane. The dotted line indicates the separation of the probe beam by typical DIC prisms applied in a commercial microscope.

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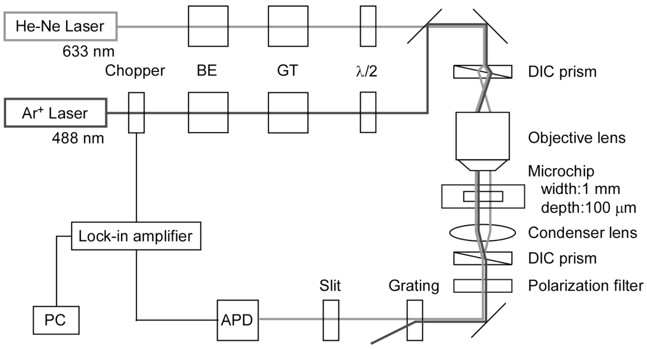

Figure 2. Schematic of the DIC-TLM: BE, beam expander; GT, Glan-Thomson prism; $\lambda/2$, half-wave plate; APD, avalanche photodiode.

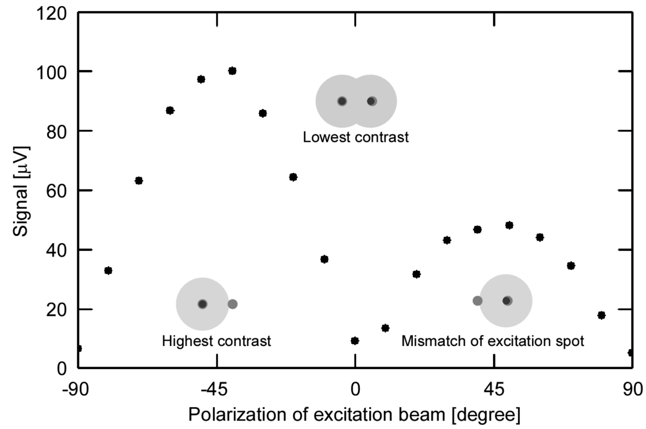

Figure 3. Dependence of DIC-TLM signal on the polarization of the excitation beam. The figures of beam spots and thermal lens show the phase contrast when polarizations are −45, 0, and +45 degrees, respectively.

is approximately 0.5 μm. For this case, however, the thermal lens is overlapping with the two probe beam spots. This leads to low phase contrast and sensitive detection of the photothermal effect becomes impossible. In order to produce high phase contrast, the thermal diffusion length and the shear value must be adjusted. One solution for this is a high modulation frequency. If f is adjusted in order to realize $\mu = 0.5$ μm, f becomes higher than 180 kHz. However, the intensity of the photothermal signal generally decreases for higher modulation frequencies, as the generated thermal energy is inversely proportional to f. The limit at lower frequencies is determined by the residence time of the analyte in the focused area of the excitation beam. As particles pass through the focus area in several milliseconds, f should be higher than 1 kHz to realize synchronous signal processing with a lock-in amplifier. Therefore, a modulation frequency of 1 kHz will be almost the optimum frequency, and an expansion of the shear value is the better solution to realize sensitive detection of individual particles in liquid. Here, we fabricated a pair of DIC prisms with a shear value of 5 μm. The shear value was confirmed by observing the focused probe beams directly using a CCD camera with an objective lens (magnification: 20×). The distance of the two probe beam spots was approximately 5 μm as designed. The configuration of the spots and the thermal lens is illustrated in Figure 1b.

Setup. Figure 2 shows a schematic diagram of the DIC-TLM. A 488 nm argon ion laser was used as an excitation beam. The excitation beam was chopped by a light chopper at a modulation frequency of 1 kHz. A 633 nm He−Ne laser was applied as a probe beam. Both beams were linearly polarized by Glan−Thomson prisms, and their polarization planes were rotated by half-wave plates. The polarization plane of the excitation beam was adjusted to 45 degrees, and its extinction ratio was 10:1. The polarization plane of the probe beam was adjusted to 0 degrees, and its extinction ratio was 1000:1. As a result of this polarization control, the probe beam was separated 1:1 by an upper DIC prism installed in a microscope (Eclipse 80i, Nicon Corporation, Japan), while the excitation beam was separated in a ratio of 10:1, which was enough to produce phase contrast. After this, both beams were focused on a sample in a microchip by an objective lens (magnification: 40×, NA = 0.75). The microchip was made of fused-silica, and the applied microchannel was 500 μm wide and 100 μm deep. After passing through the sample, the probe beams were interfered by a lower DIC prism and a polarization filter. Finally, the excitation beam was cut by a filter, and the remaining probe beam was detected by an avalanche photodiode. Its electric signal was fed into a lock-in amplifier (5610B, NF Corporation, Japan), and the component synchronized to the chopper frequency was extracted. The signal amplitude was recorded by a PC.

Here, two measurement modes were utilized for DIC-TLM. One was a concentration determination mode, and the other was a counting mode. In the concentration determination mode, the time constant of the lock-in amplifier was set to 1 s. In addition, the number of analyte molecules passing the focus area of the excitation beam was set to more than 1. In this case, the average thermal energy generated from many molecules that enter and leave the focused area contributes to the signal which is almost constant. In the counting mode, the time constant was set to 1 ms and the number of particles in the detection volume was as low as 0.1 by diluting the solution. As particles pass through the focus area in several milliseconds, pulse signals were observed with each passage of individual particles.

RESULTS AND DISCUSSION

Verification of Principle (Concentration Determination Mode). At the start, we verified the principle of DIC-TLM. The concentration determination mode was utilized for measuring a Sunset Yellow solution (10^{-5} M). First, we investigated the background reduction effect. The intensity of the probe beam in front of the detector was compared before and after inserting the polarization filter. As a result, the intensity was reduced by 2 orders (from 20 to 0.2 μW), and a clear background reduction was observed. The signal-to-background (S/B) ratio was compared to that of conventional TLM. The S/B ratios were 4.9×10^{-2} for DIC-TLM and 5.1×10^{-3} for conventional TLM. The S/B ratio was improved by 1 order of magnitude, and the background reduction was effective for TLM measurement.

Subsequently, we verified the signal generation mechanism by rotating the polarization plane of the excitation beam. The result is shown in Figure 3. The signal changed periodically with rotating the polarization plane. A polarization of +45 and −45 degrees produced the highest phase contrast and maximum signals. Interestingly, +45 and −45 degree signals had different peak heights, which might be caused by a different shear value for the

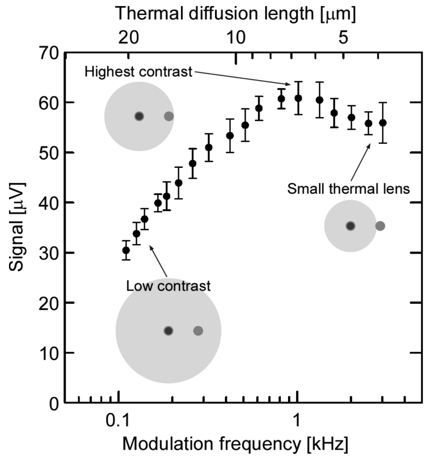

Figure 4. Dependence of DIC-TLM signal on the modulation frequency. The corresponding thermal diffusion length was calculated according to eq 1.

excitation wavelength. The polarization at 0 degrees produced the lowest phase contrast and a signal of almost zero as the same phase change was induced for the two probe beam spot. This result proves that the signals were generated only by phase contrast and refraction of the probe beam applied in conventional TLM was negligible. This is due to the low intensity of the probe beam at the detector. This result is also the first direct verification of the DIC-TLM effect in a liquid.

In a follow-up experiment, we examined the effect of the thermal diffusion length on the signal by changing the modulation frequency. The result is shown in Figure 4. The signal had maximum value at a modulation frequency of 1 kHz. This modulation frequency corresponded to a thermal diffusion length of 7 μm. This result indicates the combination of a modulation frequency of 1 kHz and a shear value of 5 μm produced the highest contrast. A higher frequency than 1 kHz led to lower sensitivity due to lower thermal energy. A lower frequency also resulted in lower sensitivity due to lower phase contrast caused by a longer thermal diffusion length.

From the three experiments, the principle of DIC-TLM was verified for the first time in a liquid. The background reduction by interference improved the S/B ratio. The signal was generated only from phase contrast. The expansion of the shear value (from 0.5 to 5 μm) was essential to realize high sensitivity.

Spatial resolution is briefly discussed. Previously, we determined in-plane resolution of conventional TLM in a scanning measurement. According to this study, the in-plane resolution of TLM is determined by the spot size of the excitation beam and is independent of thermal diffusion length.[12] This is because that thermal lens signal is dominated by absorbance of sample in the focal volume in a scanning measurement. The same will be true about DIC-TLM. A thermal diffusion length and the shear value would not affect in-plane resolution. On the other hand, *z*-axis resolution of DIC-TLM is more complicated. The resolution might be determined by the overlap of the thermal diffusion area and the spot of the probe beam which is completely different with TLM. Shear value will affect the *z*-axis resolution.

Nanoparticle Counting (Counting Mode). Figure 5 shows the result of gold nanoparticle (5 nm in diameter) counting by conventional TLM and DIC-TLM. In the result, many pulse signals were observed. In order to verify that the pulse signals were generated by the photothermal effect, we confirmed that the pulse signals were detected only when both excitation and probe beam were irradiated. Figure 6 shows the calibration curve for pulse numbers obtained with DIC-TLM. It shows good linearity in

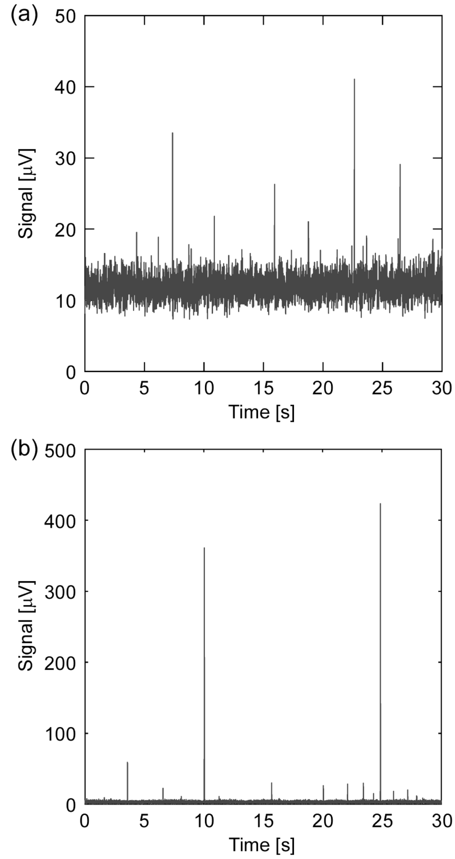

Figure 5. Counting of gold nanoparticles (5 nm in diameter) by (a) conventional TLM and (b) DIC-TLM. The average excitation power was 10 MW cm^{-2}. The concentrations were 100 pM in (a) and 83 pM in (b). The maximum signal-to-noise (S/N) ratios of the pulse signals were 7 in (a) and 70 in (b).

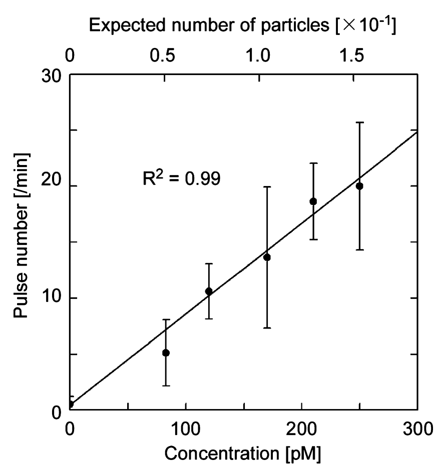

Figure 6. Calibration curve for gold nanoparticle counting. Each measurement was conducted for 1 min and repeated 10 times. The average values and standard deviations of the pulse numbers are presented. The expected number of particles was calculated from the concentration and detection volume. In our experiment, we assumed a cylindrical detection volume with a diameter of the excitation beam spot size (800 nm) and a height of one confocal length (1 μm).

relation to the concentration of particles. These results show that the pulse signals were photothermal signals resulting from the passage of individual particles. The important point of these measurements is that signal-to-noise (S/N) ratio for the detection of pulse signals was improved by 1 order (from 7 to 70) in DIC-TLM. This is due to the background-free photodetection and the resulting low fluctuation of the probe beam at the modulation frequency. No other method has shown such high sensitivity for the detection of metallic nanoparticles. Hence, this method is expected to be applied in single molecule detection of nonfluorescent molecules. As an example, we discuss the case of molecules with a molar absorption coefficient of 10^5 M^{-1} cm^{-1}. The absorption cross section is 2 orders of magnitude smaller compared to that of gold nanoparticles. In addition, molecules have optical saturation effects due to the long thermal relaxation time. According to our investigation, the photothermal signal of a molecule decreases by 1 order of magnitude by this effect. The decreased sensitivity can be recovered by an organic solvent to some extent. Chloroform, for example, enhances the photothermal effect 30-fold, owing to its low thermal conductivity and high temperature gradient of refractive index.[19] Considering all these factors, the S/N ratio for single molecule counting with DIC-TLM is estimated to be 2 to 3. Following the above discussion, the performance of DIC-TLM reaches to a single molecule.

CONCLUSION

A DIC-TLM was developed to realize background-free photodetection in a liquid. A pair of DIC prisms with a large shear value of 5 μm was fabricated to produce high phase contrast. The principle of DIC-TLM was verified by three experiments. First, the background was reduced to 1/100 by interference, and the S/B ratio was improved by 1 order of magnitude. Following this, the signal was proven to originate solely from phase contrast. In the third experiment, the signal reached a maximum value at a modulation frequency of 1 kHz, which indicated that the expansion of the shear value was effective. Subsequently, we demonstrated counting of gold nanoparticles (5 nm in diameter) in a liquid. Pulse signals of individual particles were detected, and the S/N ratio was improved by 1 order of magnitude compared to conventional TLM. The sensitivity of the DIC-TLM was estimated to be high enough to detect individual molecules, considering the molecule's absorption cross section and an enhancement of the photothermal signal by an organic solvent.

In the near future, DIC-TLM will become an innovative tool for the detection of individual molecules in liquid. For example, DIC-TLM will contribute to proteomics by detecting individual proteins without degrading their functionality by labeling.

ACKNOWLEDGMENT

This work was supported by JSPS Grant-in-Aid for Specially Promoted Research (21000007).

Received for review August 21, 2009. Accepted October 17, 2009.

AC901888U